# Diego De Pablo

**Bioinformatics Engineer | Software, Infrastructure & Data Science**

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## Professional Summary

B.Sc. in Health Engineering (Bioinformatics) and M.Sc. student in Computer Engineering. Specialized in backend engineering, data architectures, distributed systems, and applied AI, with experience in real-time telemetry and edge devices. Focused on delivering technical rigor, scalability, and continuous learning to high-impact projects.

## Professional Experience

### Software Engineer, Khaos Research — Sep 2025 – Present

*R&D on European federated data spaces (SEDIA | Gaia-X)*

- Architected and developed the Federated Data Space Ecosystem (EDAAN): automated multi-format metadata extraction into JSON-LD (DCAT-AP) with a scientific manuscript in preparation.
- Designed and integrated a hybrid search engine combining vector retrieval in **Qdrant** and lexical search, supported by knowledge-graph agents to navigate data spaces.
- Reduced Sentinel-2 satellite imagery processing time by 80% (LandCoverPy) designing a distributed pipeline (tile/season sharding) orchestrated with Dask.

### Full-Stack Developer, mSurgery — Mar 2025 – Jul 2025

*Medical telemetry platforms and real-time surgical digital twins*

- Led the frontend migration to **Svelte 5**, adopting the reactive runes paradigm.
- Delivered technical demonstrations and presentations in English to international investors and industry committees at the National Digital Content Hub.

### Junior Software Engineer, T&B Company Group — Jun 2021 – Dec 2021

*Corporate platform modernization toward REST microservices*

- Refactored legacy modules in Java and Angular toward decoupled architectures.
- Designed standard interface contracts with **OpenAPI / Swagger** and documented REST microservices to accelerate frontend-backend team integration.

## Selected Technical Projects

diegodepablo.is-a.dev/projects

### Biomedical Telemonitoring IoT Wristband (B.Sc. Thesis)

**Stack:** C++, FreeRTOS, ESP32, MQTT, FastAPI, TimescaleDB, Docker

- End-to-end telemonitoring prototype: multithreaded ESP32 firmware for vital signs via photoplethysmography (pulse and SpO2) and fall detection algorithm using MPU-6050 accelerometer.
- Wireless synchronization pipeline over MQTT to a FastAPI backend and TimescaleDB with a real-time web dashboard. **Grade:** 10/10 with Honors.

### TITAN Workflow: Scientific Workflow Orchestrator & Assistant

**Stack:** FastAPI, Pydantic AI, LangChain, LLMs, Docker

- Intelligent assistant for scientific workflow composition: validates data schemas (JSON/Avro) and stage dependencies prior to execution using Pydantic AI validators.

## Education

**M.Sc. in Computer Engineering** | *University of Malaga* — 2026 – Present

**B.Sc. in Health Engineering (Bioinformatics Specialization)** | *University of Malaga* — Sep 2021 – Jul 2025

GPA: **7.96/10**. Honors in *B.Sc. Thesis (10/10)*, *Programming Fundamentals*, *Intelligent Systems*, and *Bioinformatics Projects*.

## Technical Skills & Languages

**Programming Languages:** Python, C/C++, JavaScript, TypeScript, C#, SQL, Java, Bash, R.

**Backend & Architecture:** FastAPI, REST Microservices, Dask, OpenAPI / Swagger, Pydantic, Concurrency (AsyncIO), Node.js.

**Databases & Search:** PostgreSQL, TimescaleDB, Elasticsearch, Qdrant (Vector DB), Redis.

**AI, DevOps & Systems:** Pydantic AI, LangChain, RAG / Embeddings, LLM Agents, PyTorch, Docker, Kubernetes, Helm, Argo CD, CI/CD, Linux.

**Frontend & UI:** React, Svelte 5, Next.js, HTML5, CSS, Tailwind CSS.

**Languages: Spanish** (Native) · **English** (B2 Professional).